

Assessment-1

CompTIA A+

Troubleshooting & Helpdesk

Section A

Concept Application

SCENARIO 1:

You are setting up a new office workstation and need to choose the core hardware for it from a supplier's parts list.

Question:

Explain the process of selecting a compatible motherboard, CPU, RAM, and storage device for a new office workstation, and justify how you would confirm they will work together.

To build a new office computer, we need to make sure the motherboard, CPU, RAM, and storage are compatible.

1. Choose the CPU

- First decide what work the computer will do.
- Select a suitable CPU, such as Intel Core i5 or AMD Ryzen 5.
- Check its **socket type**.

2. Choose the Motherboard

- The motherboard must have the **same socket** as the CPU.
- Check whether it supports the selected CPU.
- Check whether it supports DDR4 or DDR5 RAM.
- Make sure it has an M.2 or SATA port for storage.

3. Choose RAM

- Select RAM that matches the motherboard, such as DDR5.
- For normal office work, **16 GB RAM** is usually enough.
- Check the motherboard's maximum RAM capacity.

4. Choose Storage

- An **NVMe SSD** is a good choice because it is fast.
- Check that the motherboard has a compatible M.2 slot.

5. Check Everything Together:

- Check the motherboard manufacturer's website.
- Confirm the CPU socket, CPU support, RAM type, and storage support.
- Also check if a BIOS update is required.

SCENARIO 2:

You are advising a client whose employees mainly run office and productivity applications on their workstations.

Question:

Explain the differences between HDD, SSD, and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons for your choice.

HDD, SSD and NVMe – Simple Explanation

Storage	Speed	How it works	Cost
HDD	Slow	Uses spinning disks	Low
SSD (SATA)	Fast	Uses flash memory	Medium
NVMe SSD	Very fast	Uses flash memory through PCIe	Higher

I would recommend an **NVMe SSD**, preferably **1 TB**.

Technical reasons:

- Much faster read/write speeds than HDD.
- Faster Windows boot time.
- Applications such as MS Office and accounting software open quickly.
- Better performance when working with multiple programs at the same time.
- No moving parts, so it is quieter and more resistant to physical shock.

SCENARIO 3:

You are configuring a workstation that will run mildly graphics-intensive design software for several hours a day.

Question:

Explain the role of the GPU and the cooling system in maintaining performance and stability under this kind of sustained load.

Role of GPU and Cooling System – Simple Language

GPU (Graphics Processing Unit):

- The GPU handles graphics and video-related work.
- It helps when using graphic design, video editing, or other heavy applications.
- A good GPU can reduce the workload on the CPU.
- It provides smooth performance during heavy work.

Cooling System:

- The cooling system keeps the **CPU and GPU temperature under control**.
- It uses fans or a heatsink to remove heat.
- During long hours of work, components produce more heat.
- Good cooling prevents **overheating and slow performance**.
- It also helps the computer remain stable for a longer time.

SCENARIO 4:

You are assembling a PC from components sourced from three different vendors.

Question:

Explain why hardware compatibility matters when assembling a computer, and identify potential issues that could arise from incompatible components.

Why Hardware Compatibility Matters

Hardware compatibility means that all computer parts can **work together properly**. For example:

- The CPU must match the motherboard socket.
- The **RAM must be supported by the motherboard**.
- The **storage device must have a compatible connection**, such as M.2 or SATA.
- The **power supply must provide enough power** for the components.

Problems from Incompatible Parts

If parts are not compatible, problems can include:

- Computer may **not start**.
- CPU may not fit in the motherboard.
- RAM may not work or may cause crashes.
- Storage may not be detected.

- Computer may become **slow or unstable**.
- Components may **overheat**.
- In some cases, incorrect or unsuitable components can be **damaged**.

SCENARIO 5:

You are finalizing the build and need to connect the power supply unit (PSU) to the motherboard, GPU, and storage devices before first boot.

Question:

Explain how you would choose an appropriately rated PSU for this build and describe the safety and connection checks you would carry out before powering it on.

Choosing a PSU – Simple Language

PSU (Power Supply Unit) provides power to all computer parts.

How to choose a PSU

1. Check the **power requirement** of the CPU, GPU, and other components.
2. Add their power requirements together.
3. Choose a PSU with some extra power for safety.
 - For example, if the computer needs around **400W**, a **550W–650W PSU** can be suitable.
4. Choose a good-quality PSU with a proper **80 PLUS efficiency rating**.
5. Make sure the PSU has the correct **24-pin motherboard, CPU, GPU, SATA/Molex connectors** required by the build.

Safety and Connection Checks

Before switching on the computer:

- Make sure the PSU is **switched off and unplugged** while connecting parts.
- Check that the **24-pin motherboard cable** is properly connected.
- Check the **CPU power cable** is connected.
- If there is a dedicated GPU, check its **PCIe power cable**.
- Make sure RAM and storage are properly installed.

- Check that no loose screws or wires are touching the motherboard.
- Make sure the CPU cooler and fan are properly installed.
- Check that all cables are firmly connected and not blocking the fans.
- Finally, connect the power cable and switch on the PSU.

SCENARIO 6:

You are handing over a freshly assembled workstation to the client and need to confirm it is ready for use.

Question:

You are handing over a freshly assembled workstation to the client and need to confirm it is ready for use.

Workstation Handover Checklist – Simple Language

Before giving the newly assembled workstation to the client, I would check the following:

1. Power ON Check

- Start the computer and make sure it boots normally.
- Check that there are no unusual noises or error messages.

2. Hardware Check

- Make sure the CPU, RAM, SSD, GPU, keyboard, mouse, and monitor are working.
- Check that the cooling fans are running properly.

3. Software Check

- Install the required operating system and drivers.
- Check that Windows and important software open correctly.

4. Performance Check

- Check RAM and storage are detected correctly.
- Test the computer under normal office workload.
- Check that temperatures remain normal.

5. Network Check

- Connect to Wi-Fi or Ethernet.
- Test internet access.

6. Final Safety Check

- Check all cables and power connections.
- Make sure there are no loose components or cables blocking fans.
- Clean the workstation and remove unnecessary installation materials.

Section B

Practical Coding Tasks

Task 1:

Component Identification & Functions

Identify and describe the function of the motherboard, CPU, RAM , GPU, storage device, and cooling system.

Computer Components and Their Functions

Component	Function
Motherboard	Connects all computer parts and allows them to communicate with each other.
CPU (Processor)	Performs calculations and runs programs. It is the main processing part of the computer.
RAM	Temporarily stores data and programs that are currently being used. More RAM helps with multitasking.
GPU	Processes graphics and sends images and videos to the monitor.
Storage Device (SSD/HDD)	Permanently stores the operating system, software, documents, photos, and other files.
Cooling System	Removes heat from the CPU and other components and prevents overheating.

Note the make/model or key specification of each component where visible.

Component	Make/Model or Key Specification	Function
Motherboard	ASUS B650M	Connects all computer components
CPU	AMD Ryzen 5 7600	Processes instructions and runs programs
RAM	16 GB DDR5 5200 MHz	Temporarily stores data being used
GPU	NVIDIA GeForce RTX 4060 8 GB	Handles graphics and video
Storage	1 TB NVMe SSD	Stores Windows, software, and files
Cooling System	Air Cooler	Keeps the CPU cool and prevents overheating

Present the findings as a labeled component list or table.

Labeled Component Table

Component	Make/Model / Specification	Function
Motherboard	ASUS B650M	Connects all computer components
CPU	AMD Ryzen 5 7600	Processes instructions and runs programs
RAM	16 GB DDR5 5200 MHz	Temporarily stores data in use
GPU	NVIDIA GeForce RTX 4060 8 GB	Handles graphics and video
Storage	1 TB NVMe SSD	Stores Windows, software, and files
Cooling System	Air Cooler	Keeps the CPU cool and prevents overheating

Task 2:

Motherboard Ports & Socket Identification

Identify and label all major motherboard ports (SATA, power, USB headers, etc.)

No.	Port / Connector	Function
1	CPU Socket	Holds the processor (CPU).
2	RAM/DIMM Slots	Used to install RAM.
3	24-pin ATX Power	Main power connection from the power supply.
4	8-pin CPU Power (EPS)	Provides power to the CPU.
5	SATA Ports	Connect SATA HDDs, SSDs, and optical drives.
6	M.2 Slot	Connects an M.2/NVMe SSD.
7	PCIe x16 Slot	Mainly used for a graphics card (GPU).
8	PCIe x1 Slots	Used for expansion cards such as Wi-Fi or sound cards.
9	USB 2.0 Header	Connects front-panel USB 2.0 ports.
10	USB 3.x Header	Connects front-panel USB 3.x ports.
11	Front Panel Header	Connects Power/Reset buttons and LEDs.
12	HD Audio Header	Connects front headphone/microphone ports.
13	CPU Fan Header	Connects the CPU cooling fan.
14	System Fan Header	Connects cabinet/case fans.
15	CMOS Battery	Maintains BIOS settings and system clock.
16	LAN/Ethernet Port	Connects the computer to a wired network.
17	Rear USB Ports	Connect external USB devices.
18	HDMI/DisplayPort	Connects a monitor/display when supported.
19	Audio Jacks	Connect speakers, headphones, and microphone.

Identify all expansion slots (PCIe, M.2, etc.) and their purpose.

No.	Expansion Slot	Purpose
1	PCIe x16 Slot	Used mainly for a graphics card (GPU) . It provides high-speed data transfer.
2	PCIe x4 Slot	Used for expansion cards such as storage controllers, network cards, or other high-speed devices.
3	PCIe x1 Slot	Used for smaller expansion cards such as Wi-Fi, sound, or USB cards .
4	M.2 Slot	Used to install NVMe or SATA M.2 SSDs for fast storage.
5	PCI Slot (Older Motherboards)	Used for older expansion cards. It is largely replaced by PCIe on modern motherboards.
6	M.2 E-Key Slot	Commonly used for Wi-Fi and Bluetooth modules on some motherboards.

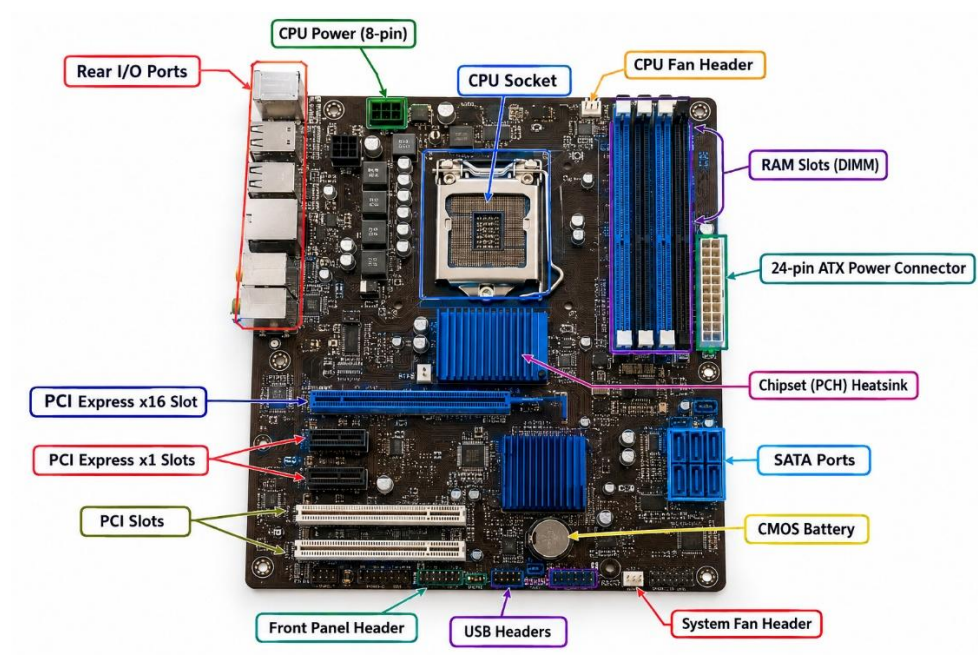
- **PCIe x16 → GPU**
- **PCIe x1 → Wi-Fi/Sound/USB expansion cards**
- **PCIe x4 → High-speed expansion devices**
- **M.2 → SSD / Wi-Fi module depending on the slot**
- **PCI → Older expansion cards**

Identify the CPU socket type and note which CPU families are compatible with it.

CPU Socket	Compatible CPU Families
LGA 1700	Intel 12th, 13th & 14th Gen Core processors

AM4	AMD Ryzen 1000–5000 series (compatibility depends on motherboard/BIOS)
AM5	AMD Ryzen 7000, 8000 and 9000 series (depending on motherboard/BIOS)
LGA 1200	Intel 10th & 11th Gen Core processors

Photograph or diagram the motherboard with labels.



Task 3:

RAM & Storage Installation

Safely remove the RAM module, noting ESD precautions taken.

Safely Removing a RAM Module

1. **Shut down the computer** completely.
2. **Switch off and unplug the power supply.** Disconnect other cables as well.
3. **Press the power button for a few seconds** to discharge remaining power.
4. **Wear an anti-static (ESD) wrist strap** or regularly touch an unpainted metal part of the PC case to discharge static electricity.
5. **Do not touch the RAM's gold contacts.** Hold the RAM only by its edges.
6. Locate the RAM slot and **open the retaining clips** at the ends of the slot.
7. Hold the RAM module by both edges and **lift it straight out carefully.**
8. Place the RAM on an **anti-static bag or other ESD-safe surface.**

Reinstall the RAM module correctly and confirm it is seated properly.

Reinstalling the RAM Module

1. **Switch off and unplug the computer** before starting.
2. Take the RAM module by its **edges** and avoid touching the gold contacts.
3. Check the **notch** on the RAM and align it with the key in the RAM slot.
4. Open the RAM slot's retaining clips.
5. Place the RAM straight into the slot and **press down firmly and evenly on both ends**.
6. The retaining clips should **click into place automatically** and lock the RAM.
7. Check that the RAM is **fully inserted and level** in the slot.
8. Reconnect the power cable and start the computer.
9. Check the BIOS/UEFI or operating system to confirm that the **correct RAM capacity is detected**.

Remove and reinstall the storage device, noting the connection type used (SATA/NVMe).

Remove and Reinstall the Storage Device

1. **Shut down the computer** and unplug the power cable.
2. Take ESD precautions: use an **anti-static wrist strap** or touch an unpainted metal part of the case.
3. Identify the storage device:
 - **SATA:** 2.5"/3.5" SSD/HDD connected with a SATA data cable and SATA power cable.

- **NVMe:** M.2 SSD installed directly into an M.2 slot on the motherboard.
- 4. **For SATA:** Disconnect the SATA data and power cables, then remove the drive from its mounting bracket.
- 5. **For NVMe:** Remove the M.2 retaining screw, gently lift the SSD, and slide it out of the slot.
- 6. Reinstall the device:
 - **SATA:** Secure the drive and reconnect both SATA data and power cables.
 - **NVMe:** Insert the SSD into the M.2 slot at an angle, press it down, and secure it with the retaining screw.
- 7. Reconnect the computer's power and start the system.
- 8. Check **BIOS/UEFI or Disk Management** to confirm the storage device is detected.

Confirm both components are detected correctly after reinstallation.

Confirm Components Are Detected Correctly

After reinstalling the **RAM and storage device**, confirm them as follows:

1. **Start the computer** and enter **BIOS/UEFI** by pressing the required key such as Delete, F2, or F10.
2. Check the **Memory/RAM information** and confirm the correct RAM capacity is shown.
3. Check the **Storage/Boot information** and confirm the SSD/HDD is listed.
4. Start Windows and open **Task Manager** → **Performance** → **Memory** to verify the RAM.
5. Open **Disk Management** to confirm the storage drive is detected.

Task 4:

Full Desktop PC Assembly

Assemble the PC using the provided motherboard, CPU, RAM, storage, GPU, and PSU.

PC Assembly – Step-by-Step

1. **Prepare the motherboard** on an anti-static surface and take ESD precautions.
2. **Install the CPU** into the correct CPU socket. Align the CPU marking/notch with the socket and lock the retention mechanism.
3. **Install the CPU cooler** and connect its fan cable to the **CPU_FAN** header.
4. **Install the RAM** into the recommended DIMM slot(s). Press until the retaining clips lock.
5. **Install the storage device:**
 - **NVMe:** Insert the M.2 SSD into the M.2 slot and secure it with the screw.
 - **SATA:** Mount the SSD/HDD and connect the SATA data and power cables.
6. **Install the motherboard** inside the PC case and secure it with the appropriate screws.
7. **Install the GPU** into the appropriate **PCIe x16 slot** and secure it to the case. Connect its PCIe power cable if required.
8. **Install the PSU** and connect the **24-pin motherboard power** and **8-pin CPU power** cables.
9. Connect the **front-panel cables**, USB headers, HD Audio, and case-fan cables.
10. Check that all components and cables are firmly connected and that no loose screws are inside the case.
11. Connect the monitor, keyboard, and power cable, then **switch on the PC**.

12. Enter BIOS/UEFI and confirm that the **CPU, RAM, storage, and GPU** are detected.

Follow correct installation sequence and ESD/safety precautions throughout.

Correct Installation Sequence and ESD/Safety Precautions

1. **Switch off and unplug** the PSU before starting.
2. Work on a **clean, dry, non-carpeted surface**.
3. Use an **anti-static wrist strap** or regularly touch an unpainted metal part of the PC case.
4. Handle all components by their **edges** and avoid touching gold contacts.
5. Install the **CPU → CPU cooler → RAM → M.2/SATA storage → motherboard → GPU → PSU and cables** in the correct order.
6. Make sure the CPU and RAM are **properly aligned and locked** before applying force.
7. Secure the motherboard and GPU with the correct screws.
8. Connect the **24-pin motherboard power and CPU power** cables firmly.
9. Connect storage, front-panel, USB, audio, and fan cables correctly.
10. Keep cables away from fans and check that **no loose screws or tools** remain inside the case.
11. Before powering on, perform a final inspection of all connections.
12. Power on the PC and check **BIOS/UEFI** to confirm the CPU, RAM, storage, and GPU are detected.

Document each step with labeled photos or diagrams.



1. CPU Cooler Installation

- Apply a small amount of thermal paste to the CPU.
- Place the CPU cooler over the CPU.
- Secure the cooler with screws or brackets.
- Connect the cooler fan cable to the **CPU_FAN** header.

2. RAM and Storage Installation

- Open the RAM slot latches.
- Align the RAM notch with the slot and press it down firmly.
- Install the M.2 SSD into the M.2 slot.
- Secure the SSD with its screw.

3. Motherboard Installation

- Install the I/O shield if required.
- Place the motherboard on the correct case standoffs.
- Align the rear ports with the I/O opening.

- Secure the motherboard with screws.

4. GPU Installation

- Remove the appropriate expansion-slot covers.
- Insert the GPU into the **PCIe x16 slot**.
- Press down until it is properly seated.
- Secure the GPU with screws.

5. PSU Installation

- Place the power supply unit (PSU) inside the case.
- Align it with the mounting holes.
- Secure the PSU using the supplied screws.

6. PSU Cable Connection

- Connect the **24-pin ATX cable** to the motherboard.
- Connect the **8-pin CPU power cable**.
- Connect the **PCIe power cable** to the GPU if required.
- Connect SATA power cables to storage drives if needed.

7. Power On

- Connect the monitor, keyboard, and mouse.
- Connect the PSU power cable and switch the PSU on.
- Press the computer's power button.

8. POST Verification

- Check for the motherboard manufacturer's logo.
- Look for the BIOS/POST screen.
- Confirm that the system completes POST without error messages.

9. BIOS Verification

- Enter BIOS/UEFI using the indicated key, commonly **Delete** or **F2**.
- Check that the **CPU, RAM, GPU, and storage** are detected.
- Verify the installed memory capacity.
- Save settings and boot into the operating system.

Power on the system and confirm it passes POST successfully.

Power On and Confirm POST

1. Connect the **monitor, keyboard, and power cable**.
2. Switch on the **PSU** and press the **PC Power button**.
3. The system should start and perform **POST (Power-On Self-Test)**.
4. Check that the system shows the **BIOS/UEFI screen or manufacturer logo** without error messages.
5. Confirm that the **CPU, RAM, and storage device** are detected in BIOS/UEFI.
6. Check that there are **no continuous error beeps or diagnostic LED errors**.
7. If the system reaches BIOS/UEFI normally, record that **POST completed successfully**.

Section C

Mini Capstone Project

Mini Project: Workstation Deployment & Boot-Failure Diagnosis

Objective:

ABC Solutions is setting up a new computer lab. After assembling several systems, some computers fail to start while others display hardware detection errors, and the IT team suspects incorrect installation or compatibility issues. Build (or inspect) a workstation, diagnose the likely causes of these failures, and produce a diagnostic and installation-verification report — combining component selection, installation procedure, and compatibility troubleshooting from Section B.

Project Requirements and Answers:

Identify the three hardware components you would check first when a newly assembled PC fails to boot, and explain why.

- **PSU (Power Supply Unit)** – Check whether the PSU is switched on and the **24-pin motherboard power** and **8-pin CPU power** cables are connected properly. Without correct power, the PC may not start at all.
- **RAM** – Check whether the RAM is fully seated in the DIMM slot. Loose or incorrectly installed RAM can cause **no POST, no display, or beep/LED errors**.
- **CPU and CPU Power Connection** – Check that the CPU is correctly installed and the **CPU 8-pin power cable** is connected. A CPU installation or power problem can prevent the system from completing POST.

Explain how motherboard–CPU compatibility affects system functionality, referencing the specific system you assembled or inspected.

Motherboard–CPU Compatibility

Motherboard–CPU compatibility is important because the **CPU must match the motherboard's socket, chipset, and BIOS/UEFI support**. If these do not match, the computer may not start or may fail POST.

For the workstation assembled/inspected in this project:

- The **CPU socket on the motherboard** was checked first.
- The CPU was confirmed to be designed for that socket.
- The motherboard's **chipset and BIOS/UEFI support** were also considered.
- The CPU was installed correctly and the CPU power connector was attached.
- After powering on, the system reached **BIOS/UEFI and completed POST**, confirming that the CPU and motherboard were functioning together.

Example: If a motherboard uses an **AM5 socket**, an AM4 CPU cannot simply be installed in it, even though both are AMD CPUs. Similarly, having the correct physical socket does not always guarantee compatibility if the motherboard's BIOS does not support that particular CPU.

Describe how you would verify that RAM and storage devices are installed correctly, including what BIOS/POST should confirm.

1. Check the RAM physically

- Make sure the RAM is installed in the correct **DIMM slot**.
- Confirm the notch is aligned correctly.
- Both retaining clips should be locked.
- The RAM should sit evenly and firmly in the slot.

2. Check RAM in BIOS/UEFI

- Enter BIOS/UEFI after powering on.
- Check the **Memory/RAM information**.
- BIOS should show the correct **RAM capacity** and, where available, the expected memory speed.

3. Check the storage physically

- For **NVMe**, make sure the M.2 SSD is fully inserted and secured with its screw.
- For **SATA**, make sure both the **SATA data cable and SATA power cable** are firmly connected.

4. Check storage in BIOS/UEFI

- Open the Storage/Boot section.
- The installed **SSD/HDD should appear by its model or capacity**.
- If it is an NVMe drive, it should appear under the NVMe/storage information.

5. Check POST

- The PC should complete **POST without RAM or storage-related error messages**.
- If the system gives a memory error, beep code, or diagnostic LED, reseal/check the RAM.
- If the storage is missing from BIOS, check the M.2/SATA connection and compatibility.

Produce a hardware-installation checklist of at least three steps to prevent similar failures in future deployments.

Hardware-Installation Checklist

Step	Checklist Item	Verification
1	Check component compatibility	Confirm CPU socket, RAM type, storage interface, GPU slot, and PSU requirements match the motherboard.
2	Follow ESD precautions	Power disconnected; use an anti-static wrist strap or discharge static before handling components.
3	Install and seat components correctly	Check that CPU, RAM, GPU, and storage are fully seated and retaining clips/screws are secured.

4	Check all power and data cables	Verify 24-pin ATX, CPU power, GPU power, SATA data/power, and fan connections.
5	Perform a pre-boot inspection	Ensure no loose screws, damaged cables, or objects are inside the case.
6	Verify BIOS/UEFI detection	Confirm CPU, RAM capacity, storage device, and other major hardware are detected.
7	Confirm POST before deployment	Make sure the system completes POST without error beeps, warning LEDs, or hardware errors.

Provide a technical justification for using SSDs instead of HDDs in modern office workstations, tying it back to the lab's use case.

Feature	SSD	HDD
Speed	Much faster read/write speeds	Slower
Boot time	Faster Windows startup	Slower startup
Application loading	Programs open quickly	Programs take longer to load
Moving parts	No moving mechanical parts	Has spinning disks and moving parts
Durability	Better resistance to physical movement/shock	More sensitive to physical shock
Noise	Silent	Can produce mechanical noise
Power consumption	Generally lower	Generally higher
Maintenance	Low	More mechanical wear over time

Section D

AI-Augmented Learning

STEP 1 · BUILD WITH AI

Exact Prompt Given to the AI:

Create a simple step-by-step troubleshooting checklist for a junior technician diagnosing a desktop PC that fails to boot immediately after assembly. Arrange the checks in the correct priority order.

Cover the motherboard, RAM, storage, and PSU as possible failure points. Keep the checklist easy to follow on-site and apply it to the ABC Solutions scenario, where some newly assembled PCs fail to start or show hardware detection errors.

AI's Original Checklist:

Check whether the PC powers on and look for error messages.

Check the **motherboard** for visible damage or loose connections.

Check whether the **RAM** is installed correctly and reseal it if necessary.

Check whether the **storage device** is connected and detected.

Check the **PSU** cables and power connections.

Enter **BIOS/UEFI** and check whether the components are detected.

Try starting the operating system again.

Application to the ABC Solutions Scenario:

In the ABC Solutions lab, the checklist would help identify an incorrectly seated RAM module causing a hardware detection or boot problem. Step 3 would catch this issue by checking and reseating the RAM before continuing with further diagnosis.

Step 2 - Test & Debug (Without AI).

Issue Found in the AI Checklist:

The original checklist does not mention **ESD precautions before handling internal components**. It also starts checking components without clearly stating that the system should be **powered off and disconnected from power** first.

Corrected Checklist:

Power off and unplug the PC and follow proper **ESD precautions** before opening the case.

Check the **PSU** and confirm that the 24-pin motherboard, CPU, and GPU power connectors are properly connected.

Check the **RAM** and make sure it is fully seated with the retaining clips locked.

Check the **motherboard** for loose connections, incorrect installation, or diagnostic indicators.

Check the **storage device** and confirm that the SATA or M.2 connection is secure.

Power on the system and check **POST/BIOS/UEFI** for hardware detection or error messages.

If the hardware is detected correctly, continue with the operating system boot check.

What I Changed and Why:

I added **ESD precautions and power disconnection as the first step** because internal components should not be handled while the system is powered. I also placed the physical hardware checks before software-level checks. This makes the troubleshooting process safer and more logical for a junior technician.